7

Evolutionary Multivariate Signals

7.1 Introduction

Some of the classical applications of chemometrics are to evolutionary data. Such type of information is increasingly common and often involves simultaneously recording spectra whilst a physical parameter such as time or pH is changed; signals evolve during the change in this parameter. Multivariate methods are useful when there are several channels or variables that are each ordered sequentially; thus, we are looking for common trends in all or some of these variables against time or any other sequential variable. The data can be organised into a multivariate data matrix, usually with the prime sequential variable represented by rows and the other variable by columns. Although the second variable may sometimes also be sequentially related (e.g. wavelength), our primary interest is to know how the process evolves over a primary variable (e.g. time), and we will call this variable (usually representing successive rows) as the *sequential variable* (Figure 7.1).

In the modern laboratory, one of the most widespread applications is in the area of coupled chromatography, such as HPLC-DAD (high-performance liquid chromatography-diode array detector), LC-MS (liquid chromatography mass spectrometry) and LC-NMR (liquid chromatography nuclear magnetic resonance). A chromatogram is recorded whilst a UV/vis, MS (mass spectrometry) or NMR (nuclear magnetic resonance) spectrum is recorded. The information can be presented in a matrix form, with time along the rows and wavelength, mass number or frequency along the columns, as already introduced in Chapter 4. Multivariate approaches can be employed to analyse these data. However, there are a wide variety of other applications ranging from pH titrations to processes that change in a systematic way with time as in kinetics. Many of the approaches in this chapter have quite a wide applicability, for example, baseline correction, data scaling and 3D PC plots; however, for brevity, we illustrate the chapter primarily with case studies from coupled chromatography, as this has been the source of a huge literature over the past decades.

With modern laboratory computers, it is possible to obtain large quantities of information very rapidly. For example, spectra sampled at 1 nm intervals over a 200 nm region can be obtained every second using modern chromatography; hence, in an hour, 3600 spectra in time × 200 spectral frequencies or 720 000 pieces of information can be produced from a single chromatogram. A typical medium to large site may contain a hundred or more coupled chromatographs, meaning the potential of acquiring 72 million data points per hour of this type of information. Add on all the other instruments, and it is not difficult to see how billions of numbers are generated on a daily basis.

In Chapters 4–6, we have discussed a number of methods for multivariate data analysis, but most of the methods described did not take into account the sequential nature of the information. When performing principal components analysis (PCA) on a data matrix, the order of the rows and columns is irrelevant. Figure 7.2 represents three cross sections through a data matrix. The first could correspond to a chromatographic peak, whereas the others not. However, as PCA and most other classical multivariate methods would not distinguish these sequences, clearly other enhancements are useful. We will primarily illustrate the methods in this chapter with reference to coupled chromatography to avoid repetition.

In many cases, underlying factors corresponding to individual compounds in a chromatographic mixture are unimodal in time, where they have one maximum. The aim is to deconvolute the experimentally observed sum into individual components and determine the features of each component. The change in spectral characteristics across the chromatographic peak can be used to provide this information.

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information that each component will be unimodal or monotonic over the course of a pH titration results in further insight. Another important application is in industrial process control where concentrations of compounds or levels of various factors may have a specific evolution over time.

A few of the methods discussed in this chapter, such as 3D PC plots and variable selection, have significant roles in most applications of chemometrics; thus, the interest in the techniques is by no means restricted to chromatographic applications; however, in order to reduce repetition, most methods are introduced in one main context. In reality, there are hundreds of bespoke methods often differing for each type of instrumental technique and problem: a full description would fill up several books. Many are very specialist or advocated by one group for one application but can be very successful and are incorporated into instrumental software. In this chapter, we will only introduce a few, but the general principles of how to approach a problem are universal. Some of the most widespread techniques are available as public domain software and as such are well publicised, though this does not always mean they are the most appropriate, simply their advocates have been best in making user-friendly software available usually at no cost.

7.2 Exploratory Data Analysis and Pre-processing

7.2.1 Baseline Correction

A first step before applying most methods in this chapter is often baseline correction. The reason for this is that most chemometric approaches look at variation above a baseline, and if this is not done, artefacts can be introduced.

Baseline correction is best performed on each variable (such as mass or wavelength) independently. There are many ways of doing this, but it is first important to identify regions of baseline and peaks, as shown in Figure 7.3, which is of an LC-MS data set. Note that the right-hand side of this tailing peak is not used, only take regions that you are confident in. Then, normally, a function is fitted to the baseline regions. This can simply involve the average or else a linear or polynomial best fit to the baseline regions. Sometimes, identifying baseline regions both before and after a peak cluster is useful; however, if the cluster is quite sharp, this is not essential, it would be hard in the case illustrated. Sometimes, the baseline is identified over the entire chromatogram (or spectrum), in other cases separately for each pack cluster or each region or window. After that, obtain a simple mathematical model that is usually multi-linear and then subtract the baseline from the entire region of interest, separately for each non-sequential variable such as wavelength. In the examples in this chapter, it is assumed that either there are no baseline problems or correction has already been performed.

1. The baseline is usually fitted to a multi-linear model to regions where there are suspected to be no peaks.
   (a) True
   (b) False

7.2.2 Principal Component-Based Plots

These are very useful for exploratory visualisation of data; for example, how many significant components are there, and which regions of the sequential data (e.g. elution time or pH) are most characteristic? However, it is important to

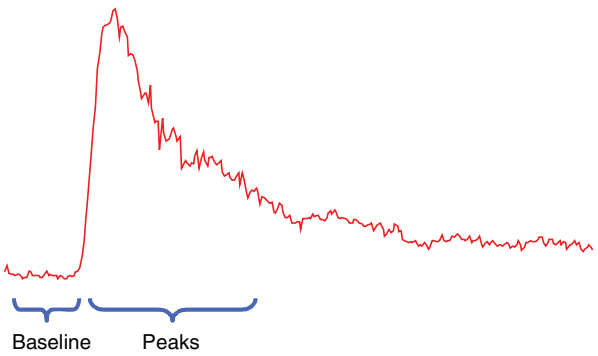


Figure 7.3 Dividing data into regions before baseline correction.

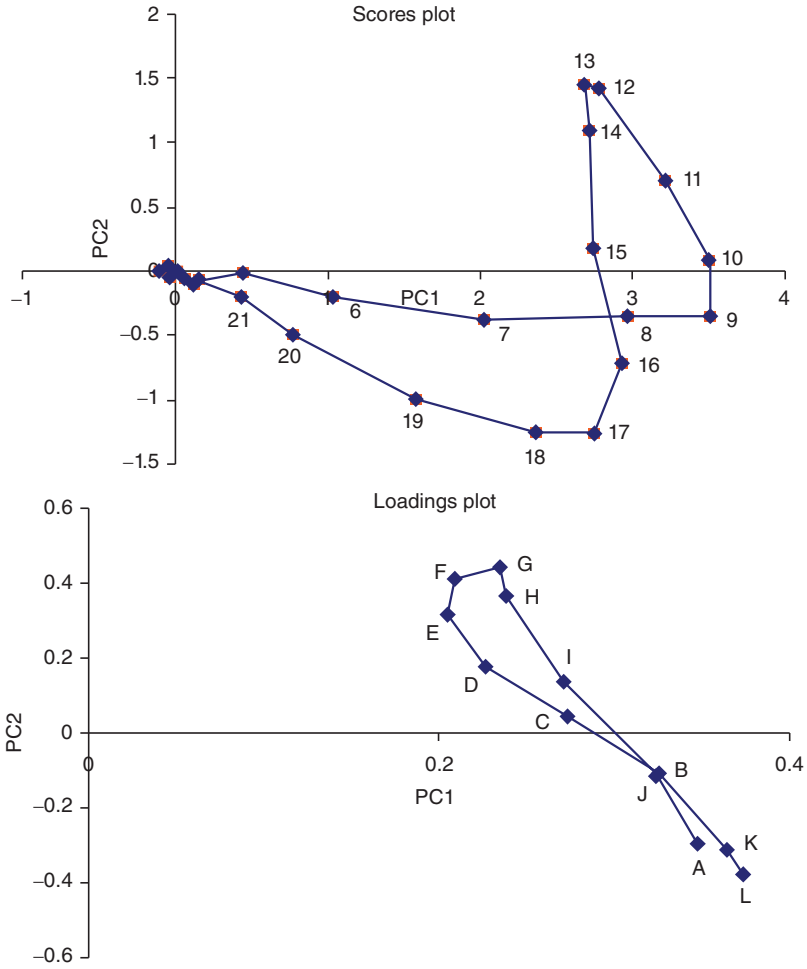


Figure 7.7 Scores and loadings plots of data set B from Table 7.2 for PC2 versus PC1.

It is sometimes clearer to present these graphs in three dimensions as shown in Figure 7.8 and Figure 7.9. Notice that the three-dimensional scores plot for data set A is not particularly informative and the two-dimensional plot shows the main trends more clearly. The reason for this is that there are only two main components in the system; hence, the third dimension consists primarily of noise and thus degrades the information. If the three dimensions were scaled according to the size of the PCs (or their eigenvalues), the graphs in Figure 7.8 would be quite flat. However, for data set B, the directions are much clearer than in the two-dimensional projections; thus, adding an extra PC can be beneficial if there are more than two significant components.

1. There is always an advantage in increasing the number of PCs from 2 to 3 in a scores plot of a hyphenated chromatogram.
   (a) True
   (b) False

7.2.3 Scaling the Data after PCA

Occasionally it is useful to scale the data after PCA, if circumstances are appropriate.

A useful trick is to normalise the scores of each PC. This involves calculating

$$^{norm}t_{ia} = \frac{t_{ia}}{\sqrt{\sum_{a=1}^{A} t_{ia}^2}}$$

in the normalised scores plot. Hence, lines are transformed to close clusters via this procedure. Obviously, if there are, for example, three significant components, it is best to transform into a sphere, or to take a small region for which it is suspected that there are only two significant components. Removing areas of noise can also be of advantage as they could appear almost anywhere.

Similar observations could be made, for example, about studying a pH titration where the different species are observed using UV/vis spectra, pH ranges consisting primarily of one species will cluster in the normalised PC scores plots and is a good way of determining what ranges correspond to just one species.

1. Several sequential points in a 2D PC scores plot fall approximately on a straight line from the origin. The PC scores are then normalised.
   - **(a)** These points will fall on an extended arc on the normalised plot in sequence.
   - **(b)** These points will fall on a straight line on the normalised plot.
   - **(c)** These points will cluster closely apart from those of low intensity.
2. If scores are normalised over two PCs, the original points will fall on an arc and be in sequence of time (or pH as appropriate).
   - **(a)** True
   - **(b)** False

7.2.4 Scaling the Data before PCA

It is also possible to scale the raw data before performing PCA. Some such techniques have been discussed in Chapter 4, but we will especially describe the application to evolutionary data such as coupled chromatography where we expect the sequential variable to consist of a series of unimodal peaks and will illustrate this by case studies A and B, discussed above, as appropriate. It is important to realise that similar principles apply, for example, in the UV/vis spectroscopy of processes such as equilibria where peaks, corresponding, for example, to the appearance and disappearance of different chemical species also have characteristic profiles along this dimension.

7.2.4.1 Scaling the Rows

Each successive row in a data matrix formed from a coupled chromatogram corresponds to a spectrum taken at a given elution time. One of the simplest methods of scaling involves summing each row to a constant total. Put mathematically

$$ {}^{const}x_{ij} = \frac{x_{ij}}{\sum_{j=1}^{J} x_{ij}} $$

Note that some people also call this *normalisation*, but we will avoid that terminology, to avoid confusion with other uses of the word. The influence on PC scores plots has already been introduced (Section 4.6.2) but will be examined in more detail in this chapter. This method introduces what is called *closure*, and we are really now just looking at how the ratio between absorbance at each wavelength changes with time. Figure 7.11(a) shows what happens if the rows of data set A are first scaled to a constant total and then PCA performed on this data. At first glance, this appears rather discouraging because the noise points have a disproportionate influence. These points contain largely nonsensical data, which are emphasised when scaling each point in time to the same total. An expansion of points 5–19 is slightly more encouraging (Figure 7.11(b)), but still not very good. Performing PCA only on points 5–19 (after scaling the rows as described above), however, provides a very clear picture of what is happening, all the points fall roughly on a straight line, with the purest points at the ends (Figure 7.11(c)). Unlike normalising the scores after PCA (Section 7.2.2), where the data must fall exactly on a geometric figure such as a circle or a sphere (depending on the number of PCs chosen), the straight line is only approximate and depends on primarily two components in the region of the data that has been chosen. It is a valuable technique for looking at mixtures spectroscopically with the position along the straight line relating to the proportion of each compound in the mixture. A little noise will result in small deviations from the straight line.

others to a total of 1? Sometimes, more than one type of measurement can be used to study an evolutionary process, such as UV/vis and MS, each data block could be scaled to a constant total. When doing selective summation, it is important to very carefully consider the consequences of pre-processing.

- *Scaling to a base peak*. In some forms of measurement, such as mass spectrometry (e.g. LC-MS or GC-MS), it is possible to select a base peak and scale to this; for example, if the aim is to analyse the LC-MS of two isomers, rationing to the molecular ion can be performed, so that

$$^{scaled}x_{ij} = \frac{x_{ij}}{x_{i(\text{molecular ion})}}$$

In certain cases, the molecular ion can then be discarded. This method of pre-processing can be used to investigate how the ratio of fragment ions varies across a cluster.

1. A series of spectra correspond primarily to a mixture of two components.
   - **(a)** If the PCs are normalised after PCA, those spectra primarily corresponding to pure compounds will fall on the extreme ends of a straight line in the scores plot.
   - **(b)** If the data are row scaled before PCA, those spectra primarily corresponding to pure compounds will fall on the extreme ends of a straight line in the scores plot.

7.2.4.2 Scaling the Columns

In many cases, it is useful to transform the columns, for example, each wavelength or mass number or spectral frequency. This can be used to put all the variables on a similar scale.

Mean centring, involving subtracting the mean of each column, is the simplest method. Many PC packages do this automatically, but may be inappropriate in the case of signal analysis because the interest is about variability above the baseline rather that around an average.

Standardisation is a common technique that has already been discussed (Section 4.6.4) and is sometimes called *auto-scaling*. It can be mathematically described by

$$^{stand}x_{ij} = \frac{x_{ij} - \bar{x}_j}{\sqrt{\sum_{i=1}^{I}(x_{ij} - \bar{x}_j)^2 / I}}$$

where there are I points in time (or length of a column), and $\bar{x}_j$ is the average of variable j. Notice that it is conventional to divide by I rather than $I-1$ in this application, if doing the calculations using many statistical packages, use the 'population' rather than 'sample' standard deviation. Matlab users should be careful when performing this scaling to ensure that the population standardisation is performed. This can be quite useful, for example, in mass spectrometry where the variation of an intense peak (such as a molecular ion of isomers) is no more significant than that of a much less intense peak, such as a significant fragment ion. However, standardisation will also emphasise variables that are pure noise, and if there are, for example, 200 mass numbers of which 180 correspond to noise, this could substantially degrade the analysis.

The most dramatic change is normally to the loadings plot. Figure 7.13 illustrates this for data set B. The scores plot hardly changes in appearance, although the centre is now shifted around the origin compared with Figure 7.7. The loadings plot, however, has changed considerably in appearance and is much clearer and more spread out.

Standardisation is most useful if the magnitudes of the variables are very different, as might occur in LC-MS. Table 7.3 is of data set C, which consists of 25 points in time and 8 variables, making a 25×8 matrix. As can be seen, the magnitude of the variables is quite different, with variable H having a maximum of 100, but others being much smaller. We assume that the variables are not in a particular sequence, or are not best represented sequentially; hence, the loadings graphs will consist of a series of points that are not joined up. Figure 7.14 is of the raw profile together with scores and loadings plots. The scores plot suggests that there are two components in the mixture, but the loadings are not very well distinguished and are dominated by variable H. Standardisation (Figure 7.15) largely retains the pattern in the scores plot, but the loadings change radically in appearance, and in this case fall approximately on a circle because there are two main components in the mixture. The variables most corresponding to each pure component fall at the

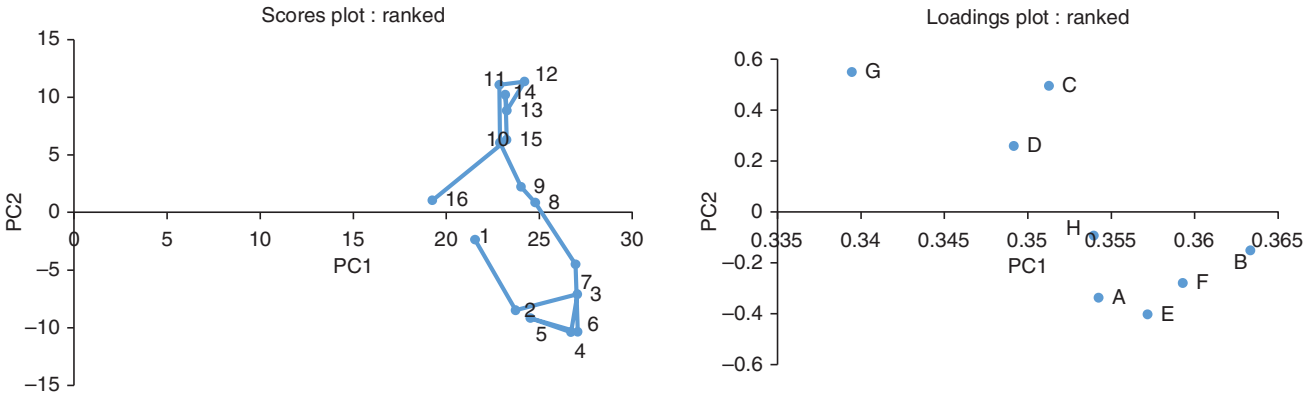


Figure 7.16 Scores and loadings of the ranked data in Table 7.4.

The PC scores and loadings plots are presented in Figure 7.16. Many similar conclusions can be deduced, as shown in Figure 7.15. For example, the loadings arising from variable *C* are close to the slowest eluting peak centred on times 14–16, whereas variables *A* to *F* correspond mainly to the fastest eluting peak. When ranking variables, it is unlikely that the resultant scores and loadings plots will fall onto a smooth geometric figure such as a circle or a line. However, this procedure can be useful for exploratory graphical analysis, especially if the data set is fairly complex with several different compounds and also many measurements on different intensity scale.

It is, of course, possible to scale both the rows and columns simultaneously, first by scaling the rows and then the columns. Note that the reverse (scaling the columns first) is rarely useful and standardisation followed by summing to a constant total has no physical meaning, as standardisation introduces negative numbers, which make row scaling inappropriate.

1. It is desired to row scale to a constant total and standardise the columns.
   - **(a)** Row scaling must be done first.
   - **(b)** Standardisation must be done first.
   - **(c)** It does not matter.

7.2.5 Variable Selection

Variable selection has an important role throughout chemometrics but will be described below primarily in the context of coupled chromatography. This involves keeping only a portion of the original measurements, selecting only those such as wavelengths or masses that are most relevant to the underlying problem. There are a huge number of combinations of approaches limited only by imagination. In this section, we give only a brief summary of some of the main methods. Often, several steps are combined.

Variable selection is particularly important in techniques such as LC-MS or GC-MS. Raw data form is sometimes called a *sparse data matrix*, in which the majority of data points are zero or represent noise. In fact, only a small percentage (perhaps 5% or less) of the measurements are of any interest. The trouble with this is that if multivariate methods are applied to the raw data, often the results are nonsense, dominated by noise. Consider the case of recording the LC-MS of two closely eluting isomers, whose fragment ions are of principal interest. The most intense peak might be the molecular ion; however, in order to study the fragmentation ions, a method such as standardisation described above is required to place equal significance on all the ions. Unfortunately, not only are perhaps 20 or so fragment ions increased in importance, but so are 200 or so ions that represent pure noise, thus the data get worse not better. Typically, out of 200 or 300 masses, there may be around 20 significant ones, and the aim of variable selection is to find these. However, too much variable reduction has the disadvantage that the dimensions of the multivariate matrices are reduced. It is important to find an optimum size as illustrated in Figure 7.17. What tricks can we use to remove irrelevant variables?

Some simple methods, often used as an initial filter of irrelevant variables, are as follows. Note that it is often important to first perform baseline correction (Section 7.2.1).

into account; variables diagnostic for an individual compound are quite likely to vary in a smooth and predictable way, whereas those due to noise will vary in a random manner.

For each type of coupled chromatography (and indeed for any technique where chemometric methods are employed), there are often specific methods for variable selection. In some cases, such as LC-MS, this is usually a crucial first step before further analysis, whereas in the case of HPLC-DAD, it is often omitted and less essential.

1. It is intended to select variables according to the ratio of variance over mean. This often does not work well, why.
   (a) Noisy variables may have small means, so the variance over mean may be fortuitously large.
   (b) Variables that are present in high intensity throughout may have low relative variance and be rejected by mistake.

7.3 Determining Composition

After exploring data via PC plots, baseline correction, pre-processing, scaling and variable selection, as required, the next step is to normally look at the composition of different regions. Many chemometric techniques try to identify pure variables that are associated with one specific component in a mixture. In chromatography, these are usually regions in time where a single compound elutes, although measurements such as an m/z value can also be characteristic of a single compound or a peak in IR spectroscopy. Below we will concentrate primarily on methods for determining pure variables in the chromatographic direction, but many can be modified quite easily for spectroscopy. There is an enormous battery of techniques, but below we summarise the main groups of approaches.

7.3.1 Composition

The concept of composition is an important one. There are many alternative ways of expressing the same idea, that of rank being popular also, which derives from matrices: ideally, the rank of a matrix equals the number of independent components or non-zero eigenvectors.

A region of composition 0 contains no detectable compounds, one of composition 1, one detectable compound and so on. Composition 1 regions are also called *selective* or *pure regions*. A complexity arises in that because of noise, a matrix over a region of composition 1 will not necessarily be described by only one PC, and it is important to try to identify how many PCs are significant and correspond to real information. The concept of composition is an approximate idea, and a composition 1 region could better be defined as a region of a chromatogram where only one compound is observable – very small tails of interferences are often not detected once they are less than a few per cent of the main peak, as their intensity is no more than the typical noise level.

There are many cases of varying difficulty. Figure 7.18 illustrates four cases as exemplified by chromatography. Case (a) is the most studied and easiest, in which each peak has a composition 1 or selective region. Although not the hardest

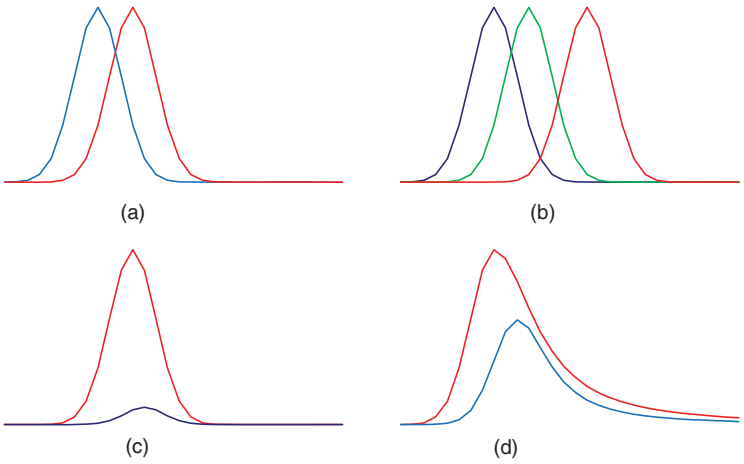


Figure 7.18 Different types of problems in chromatography.

of problems, there is often considerable value in the application of chemometric techniques in such a situation. For example, there may be a requirement for quantification in which the complete peak profile is required, including the area of each peak in the region of overlap. The spectra of the compounds might not be very clear and chemometrics can improve the quality. In complex peak clusters, it might simply be important to identify how many compounds are present, which regions are pure, so what the spectra in the selective regions are and whether it is necessary to improve the chromatography. Finally, this has a potential in the area of automation of pulling out spectra from chromatograms containing several compounds where there is some overlap. Case (b) involves a peak cluster where one or more do not have a selective region. Case (c) is of an embedded impurity peak and is surprisingly common. Many modern separations involve asymmetric peaks such as in case (d) and a lot of conventional chemometric methods fail under such circumstances.

To understand the problem, it is possible to produce a graph of ratios of intensities between the various components. For ideal peak shapes corresponding to the four cases above, these are presented in Figure 7.19. Notice the use of a logarithmic scale, as the ideal ratios will vary over a large range. Case (a) corresponds to two Gaussian peaks (for more information about peak shapes, see Section 3.2.1) and is quite straightforward. In case (b), the ratios of the first to second, and of second to third, peaks are superimposed; it can be seen that the rate of change is different for each pair of peaks; this relates to the different separation of the maxima. Note that there is a huge dynamic range, which is due to noise-free simulations being used. Case (c) is typical of an embedded peak, showing a purity maximum for the smaller component in the centre of its elution. Finally, the ratio arising from case (d) is typical of tailing peaks; many multivariate methods cannot cope easily with this type of data. However, these graphs are of ideal situations, and, in practice, it is only practicable to observe small effects if data are of an appropriate quality. In reality, measurement noise and spectral similarity limit data quality. In practice, it is only realistic to detect two (or more components) if the ratios of intensities of the two peaks are within a certain range, for example no more than 50:1, as indicated by region a in Figure 7.20. Outside these limits, it is unlikely that a second component will be detected. In addition, when the intensity of signal is sufficiently low (say 1% of the maximum, outside region b in Figure 7.20), the signal will be swamped by noise, and thus no signal is detected. Region a would appear to be of composition 2, the area of region b where region a has ended of composition 1 and the chromatogram outside region b of composition 0. If noise levels are higher, these regions become narrower.

Below we indicate a number of approaches for the determination of composition.

1. Peak B is completely embedded within a larger peak A. Over the detectable area including peak A:
   - **(a)** There will be no composition 1 region.
   - **(b)** There will be one contiguous composition 1 region.
   - **(c)** There will be two contiguous composition 1 regions.

7.3.2 Univariate Methods

So far, the simplest are univariate approaches. It is important not to overcomplicate a problem if not justified by the data. Most conventional chromatography software contains methods for estimating ratios between peak intensities at given wavelengths (or mass numbers or any non-sequential marker variable). If two spectra are sufficiently dissimilar, then this method can work well. The two most diagnostic wavelengths can be chosen by a number of means. For the data in Table 7.1, we can look at the loadings plot in Figure 7.5. At first glance, it may appear that C and G are most appropriate to use, but this is not so. The problem is that the most diagnostic wavelengths for one compound may correspond to zero or very low intensity for the other one. This would mean that there will be regions of the chromatogram where one number is close to 0 or even negative (because of noise), leading to very large or even negative ratios. Measurements that are characteristic of both compounds but exhibit distinctly different features in each case are more appropriate. Figure 7.21(a) is of the ratio of intensities of variables D to F. Initially, this plot looks slightly discouraging because there are noise regions where almost any ratio could be obtained. Cutting the region down to times 5–18 (within this range, the intensities of both variables are positive) improves the situation. It is also helpful to consider the best way to display the graph because a ratio of 2:1 is no more significant to a ratio of 1:2, yet using a linear scale, there is an arbitrary asymmetry. To overcome this, either use a logarithmic scale (Figure 7.21(b)) or take the smaller of the ratio of D:F and F:D (Figure 7.21(c)).

The peak ratio plots would suggest that there is a composition 2 region starting between times 9 and 10 and finishing between times 14 and 15. There is some uncertainty about the exact start and end, largely because noise is imposed

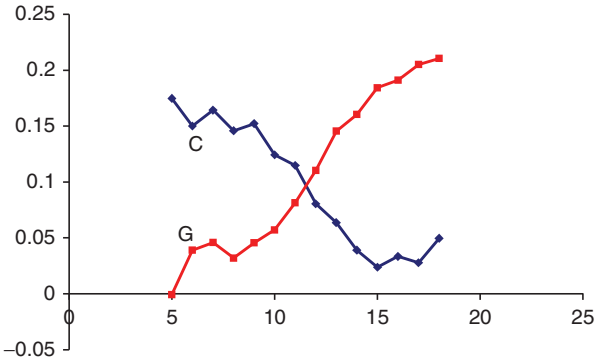


Figure 7.22 Intensities for wavelengths C and G using data of data set A summing the measurements at each successive point to constant total of 1.

Providing that noise regions are discarded, the relative intensities of diagnostic wavelengths should change according to the composition of the data; thus, we plot the value of x_{ij} for a diagnostic value of j against i. Unlike using ratio plots, we are able to choose strongly associated wavelengths such as C and G, as an intensity of 0 (or even a small negative number) will not unduly influence the appearance of the graph, given in Figure 7.22. The regions of composition 1 are somewhat flatter but influenced by noise, where the relative intensity changes most is in the composition 2 region.

This approach is not ideal, but the graphs of Figures 7.21 and 7.22 are intuitively easy for the practicing chromatographer (or spectroscopist) and result in the creation of a form of purity curve. The appearance of the curves can be enhanced by selecting variables that are least noisy and then calculating the relative intensity as a function of time for several (rather than just two) variables. Some will increase and others decrease according to whether the variable is most associated with the fastest or slowest eluting compound. Reversing the curve for one set of variables results in several superimposed purity curves, which can be averaged to give a good picture of changes over the chromatogram.

These methods can be extended to cases of embedded peaks, in which the purest point for the embedded peak does not correspond to a selective region; a weakness of using this method of ratios is that it is not always possible to determine whether a maximum (or minimum) in the purity curve is genuinely a consequence of a composition 1 region or simply the portion of the chromatogram where the concentration of one analyte is most.

Such simple approaches can become rather messy when there are several compounds in a cluster especially if the spectra are similar, but in favourable cases are very effective. Most chemometric experts advocate multivariate approaches, but traditional univariate ratio plots are still useful and often easy to understand.

1. Univariate approaches usually involve first selecting single marker variables for each compound in a mixture.
   (a) True
   (b) False

7.3.3 Correlation- and Similarity-Based Methods

Another set of methods are based on correlation coefficients. The principle is that the correlation coefficient between two successive points in time, defined by

$$r_{i-1,i} = \sum_{j=1}^{J} \frac{(x_{ij} - \bar{x}_i)(x_{i-1j} - \bar{x}_{i-1})}{s_i s_{i-1}}$$

where $\bar{x}_i$ is the mean measurement at time i and s_i the corresponding standard deviation, will have the following characteristics.

- It will be close to 1 in regions of composition 1.
- It will be close to 0 in noise regions.
- It will be substantially below 1 in regions of composition 2.

have well-established selective areas. It is possible to determine the purest point for each component in the mixture successively by extending the approach illustrated above.

Another related aspect involves using these graphs to select pure variables; this is often useful in spectroscopy, where certain masses or frequencies are most diagnostic of different components in a mixture, finding these helps the later stages of the analysis.

It is not essential to use correlation coefficients as a method for determining similarity and other measures, even of dissimilarity, can be proposed, for example, Euclidean distances between normalised spectra (scaled to unit length rather than summed to 1), the larger the distance, the less the similarity. It is also possible to start with an average or most intense spectrum in the chromatogram and determine how the spectrum at each point in time differs from this. The variations on this theme are endless and most papers or software packages contain some differences reflecting the authors' favourite approaches. When using correlation- or similarity-based methods, it is important to work carefully through the details and not accept the results as a black box. There is insufficient space in this text to itemise each published method, but the general principles should be clear for any user.

1. If two successive points are composition 1, the correlation coefficient of their successive spectra should be
   (a) Close to 0
   (b) Close to 1
   (c) Cannot say
2. A region of a chromatogram consists of a large dominant peak enveloping a smaller embedded peak in the centre. The correlation coefficients are calculated between successive spectra.
   (a) There will be one region with correlation coefficients close to 1.
   (b) There will be two regions with correlation coefficients close to 1.
   (c) There will be three regions with correlation coefficients close to 1.

7.3.4 Eigenvalue-Based Methods

Many chemometricians like as multivariate approaches mostly based on PCA. A large number of methods, such as evolving factor analysis (EFA), fixed sized window factor analysis (WFA) or heuristic evolving latent projections (HELP), among many, are available in the literature, which contain one step that involves calculating eigenvalues to determine the composition of regions of evolutionary two-way data. The principle is that the more the components in a particular region of a chromatogram, the more the number of significant eigenvalues.

There are two fundamental groups of approaches. The first involves performing PCA on an expanding window which we will call EFA.

There are several variants on this theme, but a popular one is indicated below.

Perform uncentred PCA on the first few data points of the series, for example, points 1–4. In the case of Table 7.1.

- This will involve starting with a 4×12 matrix.
- Record the first few eigenvalues of this matrix, which should be more than the number of components expected in the mixture and cannot be more than the smallest dimension of the starting matrix. We will keep four eigenvalues.
- Extend the matrix by an extra point in time to a matrix of points 1–5 in this example and repeat PCA, keeping the same number of eigenvalues as above.
- Continue until the entire data matrix is covered; thus, the final step involves performing PCA on a data matrix of dimensions 25×12 and keeping 4 eigenvalues.
- Produce a table of the eigenvalues against matrix size. In this example, there will be 22 $(=25-4+1)$ rows and 4 columns. This procedure is called *forward expanding factor analysis.*
- Next, take a matrix at the opposite end of the data set, from points 21 to 25, to give another 4×12 matrix, and calculate the eigenvalues of this matrix.
- Expand the matrix backwards; thus, the second calculation is from points 20 to 25, the third from points 19 to 25 and so on.
- Produce a table similar to that for forward expanding factor analysis. This procedure is called *backward expanding factor analysis.*

of baseline; hence, they will not always give clear answers; however, reducing these will improve the appearance of the eigenvalue plots significantly.

1. The ratio of first to second eigenvalue approximates to the ratio of intensities of the two most significant compounds in a window.
   (a) True
   (b) False
2. A chromatogram consists of 40 points in time, and a FSW of width 3. How many points in time will be represented in the graph of eigenvalues versus time?
   (a) 34
   (b) 36
   (c) 38
   (d) 40

7.3.5 Derivatives

Finally, there are a number of approaches based on calculating derivatives. The principle is that a spectrum will not change significantly in nature during a selective or composition 1 region. Derivatives measure change, hence can exploit this.

There are a large number of approaches to incorporating information about derivatives into methods for determining composition; however, the method below, illustrated with reference to the data set of Table 7.1, is effective.

- Scale the spectra at each point in time, as described in Section 7.2.4.1 so all spectra are a constant total. For pure regions, the scaled spectra should not change in appearance.
- Calculate the first derivative at each wavelength and each point in time. Normally the Savitzky–Golay method, described in Section 3.3.1.2 can be employed. The simplest case is a 5 point quadratic first derivative, see Table 3.6, so that

$$\Delta_{ij} = \frac{dx_{ij}}{di} \approx (-2x_{(i-2)j} - x_{(i-1)j} + x_{(i+1)j} + 2x_{(i+2)j})/10$$

- Note that it is not always appropriate to choose a 5 point window, this depends very much on the nature of the raw data.
- The closer the magnitude of this is to 0, the more likely the point represents a pure region, hence, it is best to use the absolute value of the derivative, $|\Delta_{ij}|$. Sometimes there are points in time at the end of a region that represent noise and will dominate the overall average derivative calculation, these points may be discarded, often this can simply be done by removing points whose average intensity is below a given threshold.
- If the variables are fairly different in magnitude, it is first useful to scale each variable to a similar magnitude, but setting the sum of each column (or variable) to a constant total

$${}^{const}\Delta_{ij} = |\Delta_{ij}| / \sum_{i=w-1}^{I-w+1} |\Delta_{ij}|$$

  where | indicates an absolute value, and the window size for the derivatives equals w. This step is optional.
- The final step involves averaging the values obtained in one of the steps discussed above (as appropriate) as follows:

$$d_i = \sum_{j=1}^{J} {}^{const}\Delta_{ij}/J \quad \text{or} \quad d_i = \sum_{j=1}^{J} |\Delta_{ij}|/J$$

The calculation illustrated for the data in Table 7.8(a) is of the data where each row is summed to a constant total. Notice that the first and last rows contain some large numbers because the absolute intensity is low, with several negative as well as positive numbers that are similar in magnitude. Table 7.8(b) is of the absolute value of the first derivatives. Notice the very large and not very meaningful numerical values at times 3 and 23. Row 22 also contains a fairly large number and thus is not very diagnostic. In Table 7.8(c), points 3, 22 and 23 have been discarded, and the

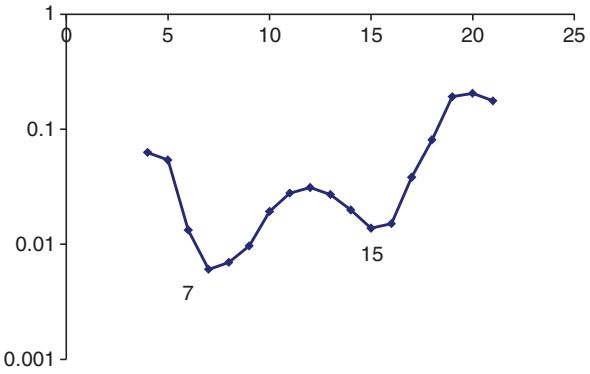


Figure 7.28 Derivative purity plot for data set A with purest points indicated.

columns have now been set to a constant total of 1. Finally, the consensus absolute value of the derivative is presented in Table 7.8(d).

The resultant value of d_i is best presented on a logarithmic scale, as shown in Figure 7.28. The regions of highest purity can be quite well pinpointed as minima at times 7 and 15. These graphs are most useful for determining the purest points in time rather than regions of differing composition, but the visual display is often very informative and can cope well with unusual peak shapes.

Of course, the derivative method is not restricted to coupled chromatography and can be used for any evolutionary process.

1. Which statement is true?
   - **(a)** If the successive spectra in time do not change significantly in appearance, then if scaled to the same total of 1, their derivatives over time at each wavelength will be close to 0.
   - **(b)** If successive spectra in time do not change significantly in appearance, then if scaled to the same total of 1, their derivatives over time at each wavelength will be close to 1.

7.4 Resolution

Resolution or deconvolution of two-way chromatograms or mixture spectra involves converting a cluster of peaks into its constituent parts, each ideally representing a component of the signal from a single compound. The number of named methods in the literature is enormous, and it would be out of the scope of this text to discuss each approach in detail. In areas such as chemical pattern recognition or calibration, certain generic approaches are accepted as part of an overall strategy, and the data pre-processing, variable selection and so on are regarded as extra steps. In the field of resolution of evolutionary multivariate data, there is a fondness for packaging a series of steps into a named method; hence, there are probably 50 or more named methods and maybe many unnamed approaches reported in the literature. Yet, most are based on a number of generic principles, which are described in this chapter.

There are several aims for resolution.

- Obtaining the profiles for each resolved compound. These might be the elution profiles (in chromatography) or the concentration distribution in a series of compounds (in spectroscopy of mixtures) or the pH profiles of different chemical species.
- Obtaining the spectra of each pure compound. This allows identification or library searching. In some cases, this procedure merely uses the multivariate signals to improve the quality of the individual spectra that may be noisy, but in other cases, such as an embedded peak, genuinely difficult information can be gleaned. This is particularly useful in impurity monitoring.
- Obtaining quantitative information. This involves using the resolved two-way data to provide concentrations (or relative concentrations when pure standards are not available).

- Automation. Complex chromatograms may consist of 50 or more peaks, some of which will be noisy and overlapping. Speeding up procedures, for example, using rapid chromatography in a matter of minutes resulting in considerable overlap, rather than taking half an hour per chromatogram, also results in embedded peaks. Chemometrics can ideally pull out the constituents' spectra and profiles.

The methods in this chapter differ from those of Chapter 6 in that pure standards are not required for the model.

Although some data sets can be very complicated, it is normal to divide the data into small regions where there are signals from only a few components. Even in spectroscopy of mixtures, in many cases such as MIR or NMR, it is normally easy to find regions of the spectra where only two or three compounds at the most absorb; thus, this process of finding windows rather than analysing an entire data set in one go is quite normal. Hence, we will limit the discussion to three peak clusters in this section. Naturally, the methods of Section 7.3 would usually first be applied to the entire data set to identify these regions. We will illustrate the discussion below primarily in the context of coupled chromatography, although it is also very well established in other areas such as equilibrium profiling.

There are several cases of increasing complexity.

7.4.1 Selectivity for All Components

The simplest problem is that all compounds (or chemical components) have some region of selectivity. These methods involve first finding some pure or selective (composition 1) region in the chromatogram or selective spectral measurement such as a m/z value for each compound in a mixture.

7.4.1.1 Pure Spectra and Selective Variables

The most straightforward situation is when each compound has an identifiable composition 1 region (often called *selective regions* in chromatography). The simplest approach is to estimate the pure spectrum in such a region. There are several methods.

- Take the spectrum at the point of maximum purity for each compound.
- Average the spectra for each compound over each composition 1 region.
- Perform PCA over each composition 1 region separately (so if there are three compounds, perform three independent PCA calculations) and then take the loadings of the first PC as an estimate of the pure spectrum. PCA is used as a smoothing technique, the idea being that the noise is banished to later PCs.

Some rather elaborate multivariate methods that use the profiles are also available, instead of using the spectra in the composition 1 regions. In the case of data set A in Table 7.1, we might guess that the fastest eluting compound A has a composition 1 region between points 4 and 8 and the slowest eluting compound B between points 15 and 19. Hence, we could divide the chromatogram as follows.

- Points 1–3: no compounds elute.
- Points 4–8: compound A elutes selectively.
- Points 9–14: co-elution.
- Points 15–19: compound B elutes selectively.
- Points 20–25: no compounds elute.

As discussed above, there can be slight variations on this theme. This is represented in Figure 7.29. Chemometrics is used to fill in the remaining pieces of the jigsaw. The only unknowns are the elution profiles in the composition 2 regions. The profiles in the composition 1 regions can be estimated either by using the summed profiles in these regions or by performing PCA in these regions and taking the scores of the first PC.

An alternative is to find pure variables rather than composition 1 regions. These methods are popular when using various types of spectroscopy such as in LC-MS or in the MIR of mixtures. Wavelengths, frequencies or masses belonging single compounds can often be identified. In the case of data set C in Table 7.3, we suspect that variables C and F belong to the two compounds (see Figure 7.15), and their profiles are presented in Figure 7.30. Note that these profiles are somewhat noisy. This is quite common in techniques such as mass spectrometry. It is possible to improve the quality of the profiles by using methods for smoothing, as described in Chapter 3, or to average profiles from several pure variables. The latter technique is useful in NMR or IR, where a peak might be defined by several data points or where there could be a number of selective regions in the spectrum.

The result of this section will be to produce either a first guess of all or part of the concentration profiles, represented by the matrix $\hat{\boldsymbol{C}}$ or of the spectra $\hat{\boldsymbol{S}}$, where the '^' or hat sign implies estimated. Although conceptually the easiest

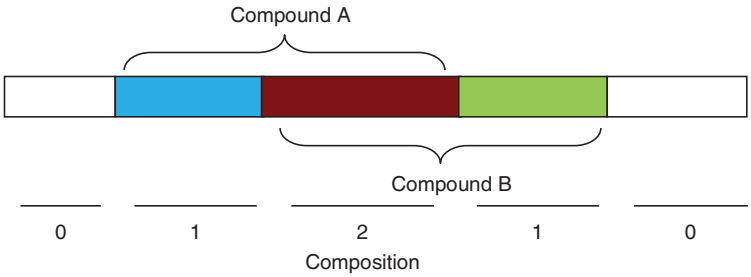


Figure 7.29 Composition of regions in chromatogram deriving from data set A.

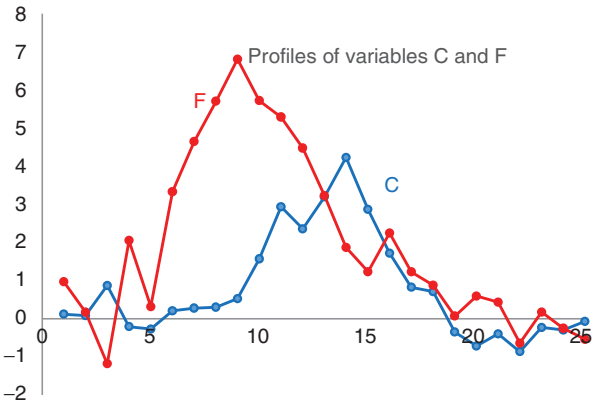


Figure 7.30 Profiles of variables *C* and *F* in Table 7.3.

approach to resolution is univariate, we see below that by using more variables, we can obtain better estimates using multiple linear regression (MLR).

1. The scores of the first principal component over a composition 1 region are an approximation of the profile of a pure compound.
   (a) True
   (b) False

7.4.1.2 Multiple Linear Regression

If pure profiles can be obtained from all components in a mixture, the next step in deconvolution is quite straightforward.

In the case of Table 7.1, we can guess the pure spectrum for A as the average of the data between times 4 and 8 and for B as the average of the data between times 15 and 19. These make up a 2×12 data matrix $\widehat{\boldsymbol{S}}$. As

$$\boldsymbol{X} \approx \widehat{\boldsymbol{C}}\widehat{\boldsymbol{S}}$$

therefore

$$\widehat{\boldsymbol{C}} = \boldsymbol{X}\widehat{\boldsymbol{S}}'(\widehat{\boldsymbol{S}}\,\widehat{\boldsymbol{S}}')^{-1}$$

as discussed in Section 6.3. The estimated spectra are tabulated in Table 7.9 and the resultant profiles are presented in Figure 7.31. Notice that the vertical scale in fact has no direct physical meaning: intensity data can only be reconstructed by multiplying the profiles by the spectra. However, MLR has provided a very satisfactory estimate, and provided pure regions are available for each significant component, it is probably entirely adequate as a tool in many cases.

If pure variables such as spectral frequencies or m/z values can be determined, even if there are embedded peaks, it is also possible to use these to obtain first estimates of elution profiles, $\widehat{\boldsymbol{C}}$, then the spectra can be obtained using all (or a great proportion of) the variables by

$$\widehat{\boldsymbol{S}} = (\widehat{\boldsymbol{C}}'\widehat{\boldsymbol{C}})^{-1}\widehat{\boldsymbol{C}}'\boldsymbol{X}$$

Table 7.9 Estimated spectra obtained from the composition 1 regions in data set A.

A	B	C	D	E	F	G	H	I	J	K	L
0.519	0.746	0.862	0.713	0.454	0.341	0.194	0.176	0.312	0.410	0.465	0.404
0.041	0.006	0.087	0.221	0.356	0.603	0.676	0.575	0.395	0.199	0.136	0.162

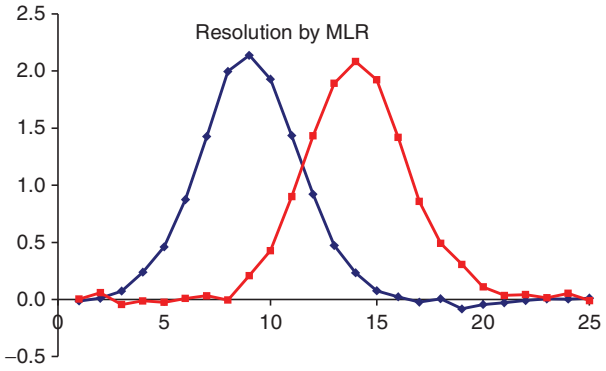


Figure 7.31 Reconstructed profiles for data set A using MLR.

The concentration profiles can be improved by increasing the number of variables; hence, for example, the first guess might involve using 1 variable per compound, the next 20 significant variables and the final 100 or more. This approach is also useful in spectroscopy of mixtures, if selective wavelengths can be identified for each compound. Using these for initial estimates of the concentrations of each compound in each spectrum, the full spectra can be reconstructed even when there are overlapping regions. Such approaches are useful in MIR, but not so valuable in NIR or UV/vis spectroscopy where it is often hard to find selective wavelengths and the effectiveness depends on the type of spectroscopy employed.

1. If a mixture consists of three components all of which have composition regions and is measured over 50 wavelengths, what are the dimensions of $\hat{\boldsymbol{S}}$?
   - **(a)** 3×50
   - **(b)** 50×3
   - **(c)** We would need to know the dimensions of the selective regions.
2. Every component has to have at least one composition 1 region for MLR to be effective.
   - **(a)** True
   - **(b)** False

7.4.1.3 Principal Component Regression

Principal component regression (PCR) is an alternative to MLR (Section 5.4) and can be used in signal analysis just as in calibration. There are a number of ways of employing PCA, but a simple approach is to note that the scores and loadings can be related to the concentration profile and spectra by

$$\boldsymbol{X} \approx \hat{\boldsymbol{C}}\,\hat{\boldsymbol{S}} = \boldsymbol{T}\boldsymbol{R}\boldsymbol{R}^{-1}\boldsymbol{P}$$

hence,

$$\hat{\boldsymbol{C}} = \boldsymbol{T}\boldsymbol{R}$$

and

$$\hat{\boldsymbol{S}} = \boldsymbol{R}^{-1}\boldsymbol{P}$$

PCR can be employed in more elaborate ways using the known profiles in the composition 1 (and sometimes composition 0) regions for each compound. These methods were the basis of some of the earliest approaches to resolution of two-way chromatographic data. There are several variants and one is as follows.

Choose only those regions where one component elutes. In our example in Table 7.1:

- We will use the regions between times 4–8 and 15–19 inclusive, which involves 10 points.
- For each compound, use either the estimated profiles if the region is composition 1 or 0 if another compound elutes in this region (so a composition 1 region for compound A will be composition 0 for compound B). A matrix is obtained of size $Z \times 2$ whose columns correspond to each component where Z equals the total number of composition 1 data points. In our example, the matrix is of size 10×2, half of the values being 0 and half consisting of the profile in the composition 1 region. Call this matrix $\boldsymbol{Z}$.
- Perform PCA on the overall matrix.
- Find a matrix $\boldsymbol{R}$ such that $\boldsymbol{Z} \approx \boldsymbol{TR}$ using the known profiles obtained in step 2, simply by using regression so that $\boldsymbol{R} = (\boldsymbol{T}'\boldsymbol{T})^{-1}\boldsymbol{T}'\boldsymbol{Z}$ but including the scores *only* of the composition 1 region.
- Knowing $\boldsymbol{R}$, it is a simple matter to reconstruct the concentration profiles by including the scores over the entire data matrix as discussed above, and similarly the spectra.

The key steps in the calculation are presented in Table 7.11. Notice that the magnitude of the numbers in the matrix $\boldsymbol{R}$ differs from that presented in Table 7.10. This is simply because the magnitude of the estimates of the spectra and profiles is different and has no physical significance. The resultant profiles obtained by the multiplication $\hat{\boldsymbol{C}} = \boldsymbol{TR}$ on the entire data set are illustrated in Figure 7.32.

In straightforward cases, PCR is unnecessary and, if not carefully controlled, may provide worse results to MLR. However, for more complex systems, it can be very useful.

1. There are three compounds in a mixture, all with composition 1 regions. Fifty samples are recorded at 100 wavelengths. The matrix $\boldsymbol{R}$ has the following dimensions.
   (a) 50×3
   (b) 3×100
   (c) 3×3

7.4.2 Partial Selectivity

Somewhat harder situations occur when only some components exhibit selectivity. A common example is a completely embedded peak in HPLC-DAD. In the case of LC-MS or LC-NMR, this problem is often solved by finding pure variables; however, as UV/vis spectra are often completely overlapping, it is not always possible to treat data in this manner.

Fortunately, PCA comes to the rescue. In Chapter 6, we have discussed the different applicability of PCR and MLR. Using the former method, we stated that it was not necessary to have information about the concentration of every component in the mixture, simply a good idea of how many significant components are there. Thus, in the case of resolution of two-way data, these approaches can easily be extended. We will illustrate this using the data in Table 7.2, which correspond to three peaks, the middle one being completely embedded in the others. There are several different ways of exploiting this.

One approach uses the idea of a zero concentration window. The first step is to identify compounds that we know have selective regions and determine where they do not elute. This information may be obtained from a variety of approaches such as eigenvalue or PC plots. In this region, we expect the intensity of the data to be zero; hence, it is possible to find a vector $\boldsymbol{r}$ for each component so that

$$\boldsymbol{0} = \boldsymbol{T}_0\boldsymbol{r}$$

where $\boldsymbol{0}$ is a vector of zeros, and $\boldsymbol{T}_0$ denotes the portion of the scores in this region; normally, one excludes the region where no peaks elute and then finds the zero component region for each component. For example, if we record a cluster over 50 data points and we suspect that there are three peaks eluting between points 10 and 25, 20 and 35, and 30 and 45, then there are three $\boldsymbol{T}_0$ matrices; for the fastest eluting component, this is between points 26 and 45. Notice that the number of PCs should be made equal to the number of significant compounds in the overall mixture. There is one small problem in that one value of the vector $\boldsymbol{r}$ must be set to an arbitrary number, usually the first coefficient is set

to 1, but this does not have a serious effect on the algorithm. The equation can then be solved as follows. Separate out the contribution from the first PC to that from all the others so, setting r_1 to 1

$$\boldsymbol{T}_{0(2:K)}\boldsymbol{r}_{2:k} \approx -\boldsymbol{t}_{01}$$

so that

$$\boldsymbol{r}_{2:K} = (T'_{0(2:K)}T_{0(2:K)})^{-1}T'_{0(2:K)}\boldsymbol{t}_{01}$$

where $\boldsymbol{r}_{2:K}$ is a column vector of length $K-1$, where there are K PCs, $\boldsymbol{t}_{01}$ is the scores of the first PC over the zero concentration window and $\boldsymbol{T}_{0(2:K)}$ the scores of the remaining PCs. It is important to ensure that K equals the number of components suspected to elute within the cluster of peaks. It is possible to perform this operation on any embedded peaks because these also exhibit zero composition regions.

The profiles of all the compounds can now be obtained over the entire region by $\widehat{\boldsymbol{C}} = \boldsymbol{TR}$ and the spectra by $\widehat{\boldsymbol{S}} = \boldsymbol{R}^{-1}\boldsymbol{P}$.

From inspecting Table 7.2, we might conclude that compound A elutes between times 4 and 13, B between times 9 and 17 and C between times 13 and 22. This information could be obtained by a variety of methods, as discussed in Section 7.3. Hence, the zero composition regions are as follows.

- Compound A: points 14–22.
- Compound B: points 4–8 and 18–22.
- Compound C: points 4–12.

Obviously, different approaches may identify slightly different regions. The calculation is presented in Table 7.12, for the data in Table 7.2, and the resultant profiles are presented in Figure 7.33. Of course, spectra could also be estimated this way.

Many a paper and thesis have been written about this problem, and there are a large number of modifications to this approach; however, in this chapter, we illustrate just using one of the best established approaches.

1. Compound A elutes between points 5 and 20, compound B between 15 and 35 and compound C between 30 and 45; data are recorded between points 1–50.
   (a) The zero concentration region for compound A is points 1–4 and 21–50.
   (b) The zero concentration region for compound A is points 21–45.
   (c) The zero concentration region for compound A is points 1–4 and 46–50.

7.4.3 Incorporating Constraints: ITTFA, ALS and MCR

Finally, it is important to mention another class of methods. In many cases, it is not possible to obtain a unique mathematical solution to the multivariate resolution of complex mixtures, and the problem of embedded components without selectivity, which may, for example, occur in impurity monitoring, causes difficulties while using many conventional approaches.

The methods above are non-iterative; that is, they have a single step that comes to a single solution. However, over the years, chemometricians have refined these approaches to develop iterative methods that are based on many of the principles mentioned above but do not stop at the first answer. These methods are particularly useful when there is partial selectivity, not only in chromatography but also in areas such as reaction monitoring or spectroscopy.

Common to all iterative methods is to start with a guess about some characteristic of each component. The methods described above are based on having quite good information about each component, whether it is a spectrum or composition 0 region, and if this information is poor, the methods will fall over, as there is only one chance to get it right, based on the starting information. In contrast, iterative methods are not so restrictive as to what the first guess is. If perfect knowledge is obtained about each component at the start, then this is useful, but iterative methods may then not be necessary.

There are, however, many other approaches for the estimation of the properties of each component in a mixture. One simple method is called a *needle* search. For each component in a mixture, the data point of maximum purity and the spectra at these points are taken as the starting guesses for each component, even though they may not necessarily correspond to pure variables. Statistical methods such as varimax rotation can also be used to provide first guesses,